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Patents Used in Patent Office Rejections as Indicators of Value

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ABSTRACT

This paper introduces a novel approach to measure a patent's economic value by examining whether the patent's disclosure leads to rejection of another pending US patent application. This approach considers the use of the patent by the US Patent and Trademark Office (USPTO) in office action rejections on the grounds of novelty or obviousness, as well as its citation as an X or Y reference in a European Patent Office (EPO) search report, which provides analogous information. Unlike conventional citation metrics widely employed by economists, the novel metric is arguably more closely tied to private value, as it is centered on the examiner's rejection of another patent application based on the patent in question. As this metric is not directly influenced by potential strategic behavior by patent lawyers or patent applicants, it is more directly tied to private value. This study evaluates how patents used in novelty and obviousness rejections, and comparable EPO information, correspond to common measures of private value—specifically patent renewal, the assertion of a patent in litigation, the number of independent patent claims, and classification in multiple technological subject matters. We examine rejection data over a defined time period for US patents issued from 1999 to 2007 and then link value data to these patents. A similar analysis is also conducted for EPO search reports. Our findings reveal that rejection uses, as well as X and Y EPO search citations, independently exhibit positive correlations with all of these value measurements. Moreover, when information from both the USPTO and EPO concerning a given patent is combined, further insights about that patent's value are obtained. We also find that rejection uses provide unique insights into additional measures of private value such as the patent being listed in the Orange Book or as a Standard Essential Patent. Accordingly, these rejection use metrics provide another independent tool for evaluating a patent's value.

1 | Background

Legal and economics literature is interested in a patent's citation. Legal scholars have shown an interest in analyzing patent citations as they provide valuable insights into the impact of the duty to disclose rules and the quality of the patent examination process (Sampat 2010; Kuhn 2010). While the economic literature underscores the importance of patent citations, particularly forward citations, as indicators of a patent's value, (Trajtenberg 1990; Marco 2007; Jaffe and de Rassenfosse 2019; Abrams, Akcigit, and Grennan 2019), it also supports the view that forward citations can serve as markers of knowledge diffusion in industries, with the company owning the citing

patent benefiting from information disclosed in the cited patent (Nelson et al. 2014). Studies have refined the understanding of which forward citations serve as better indicators of value (Jaffe and Trajtenberg 2002). However, it is critical to recognize that not all citations during the patent application process share the same circumstances and therefore may not provide the same legal and economic information.

This paper introduces a novel approach of measuring a patent's economic value by examining whether the patent's disclosure leads to the rejection of another pending US patent application. This approach considers the use of the patent by the US Patent and Trademark Office (USPTO) in office action

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rejections based on novelty or obviousness, as well as its inclusion in a European Patent Office (EPO) search report citation that provides analogous information about US patents. Unlike conventional citation metrics commonly employed by economists, this novel metric is arguably more closely tied to private value, as it reflects the substantive use of a patent by an examiner in rejecting another patent application. Because the novel metric is not directly linked to potential strategic behavior by patent lawyers or applicants, it provides a more direct connection to private value. This paper investigates how patents used in novelty and obviousness rejections, along with analogous EPO information, correlate with common measures of private value such as patent renewal, the assertion of a patent in litigation, the number of independent patent claims, and the number of technological classifications for the patent. This paper concludes by exploring other value metrics—such as inclusion in the Orange Book or as Standard Essential Patent. This is done to determine whether rejection uses correlate differentially than traditional citation metrics with these additional value metrics.

When filing a patent application with the USPTO, applicants are obligated throughout the patent examination process, to disclose all patents, and other information, relevant to patentability. (37 C.F.R. 1.56). This duty requires the disclosure of any “prior art” known to the applicant that it believes is material to the patent application’s claims’ novelty or nonobviousness. Failure to disclose relevant patents, or other prior art, can be deemed inequitable conduct. (*Therasense v. Becton Dickinson*, 649 F.3d 1276 (Fed. Cir. 2011)), rendering not only the claims relevant to the undisclosed prior art unenforceable, but also the whole patent and, potentially, all related patents (Cotropia 2009). Importantly, there is no penalty for overdisclosure (Cotter 2011). As a result, the duty to disclose, and the risk aversion of failing to meet this duty, prompt patent applicants to overdisclose and cite prior patents to the USPTO even if they are only remotely relevant. It is worth noting that patent attorney behavior influences citations, which can be affected by factors such as risk aversion, fee structure, or client preferences. Evidence suggests that a minority of patent applicants flood the patent office with a huge number of prior art disclosures (Kuhn, Younge, and Marco 2020). Moreover, there is no duty for applicants to search for prior patents, meaning that highly relevant prior art may remain undisclosed simply due to a lack of knowledge. In other words, if a patent applicant knows of material prior art, she has a duty to disclose it. If she does not know of a certain piece of prior art, she has no duty to investigate.

While most patent citations are provided by patent applicants, many of these citations have minimal relevance to the cited patent. The interplay between the duty to disclose and applicant citations can “dilute” the legal (relevance to the legal question of patentability) and economic (relevance to the same cumulative innovation process) signal applicant citations provide (Sampat 2010). Overdisclosure of irrelevant art, which is largely driven by patent attorney behavior rather than innovator behavior, dilutes the signal emitted from truly important forward citations. The absence of a duty to search, combined the fact that many patent applicants do not search for prior patents before filing an application, means that even if an

applicant over discloses, the applicant may simply not be in possession of the legally and economically relevant prior art. Studies have shown that applicant citations, which can be influenced by other, non-value related factors, are not reliable indicators of the economic value of a patent (Sampat 2010). Others have argued that citations reflect strategic behavior separate from importance (Arora, Bazenton, and Lee 2018). Furthermore, applicant citations are not extensively utilized by the USPTO during patent examination (Kuhn 2010; Alcácer, Gittelman, and Sampat 2009; Cotropia, Lemley, and Sampat 2013). Therefore, despite citations being relied upon by economists as markers of value, there is reason to be concerned that applicant citations may be driven by forces unrelated to the importance of the patent.

In contrast to patent applicants, patent examiners have a duty to search for prior art. Their primary role during the examination process is to conduct searches for prior art that is relevant to the patentability of the applied for claims and utilize this art in issuing rejections. Patent examiners evaluate whether a pending application meets the requirements for patentability (Cotropia 2009). In doing so, examiners review the prior art and provide written notice through office actions to applicants if the invention is deemed unpatentable. During examination, patent examiners conduct their own searches for prior art relevant to the patentability of the applied for claims. Patent examiners compare the found prior art and the applicant cited art to each of the claims of the patent application to determine if the claimed invention has been previously disclosed, rendering it not novel, or if it is obvious in light of prior disclosures, and thus obvious. Then, if the patent examiner believes one or more claims are not novel or obvious in light of these prior patents or publications, the patent examiner details their position in the form of a rejection—a rejection for lack of novelty and/or nonobviousness—that is reported in an office action sent to the patent applicant. This office action includes the rejections and the relied upon prior patents and publications the examiner believes render the pending patent application’s claims unpatentable (Cotropia and Schwartz 2020).

Unlike applicants, examiners have no obligation to disclose, and their performance is not evaluated based on the number of citations made in a given patent application. Their focus is primarily on evaluating the patentability of the application’s claims and issuing the rejections when the prior art renders a claim not novel or obvious. Unlike applicants, examiners have no incentive to over cite and instead simply examine the patent application under the relevant patentability standards. Patent examiners often cite a piece of art but not use it in a rejection.

This holds true for the EPO’s examination of patent application. The EPO conducts an initial search report on such applications, citing art that the EPO deems of particular relevance to the application. This level of relevance is determined by the EPO and denoted in a search report, with some art falling in category “A” indicating the art provides mere background on the technology described in the application. Other art is identified in the “X” category, meaning the art brings the application’s novelty in question. Art may fall in the “Y” category, indicating the art may invalidate the application due

to a lack of an inventive step (European Patent Office 2023). X category art aligns with art used in USPTO § 102, novelty rejections, while Y categorized art is similar to art used in a USPTO § 103, obviousness rejection, and these have been utilized by researchers involving European patents. (e.g., von Graevenitz, Wagner, and Harhoff 2011; Hall, Helmers, and von Graevenitz 2021). For the purposes of the present study, we focus on X and Y search report citations, which are analogous to novelty and obviousness rejection uses.

Because of the disparity in search and disclosure incentives between applicants and examiners, examiner citations are considered more legally and economically relevant to the cited patent (Hegde and Sampat 2009; Kuhn, Young, and Marco 2020). Examiner citations are also more likely to be employed in rejecting the patent application claims, establishing that the claimed invention lacks novelty or is obvious (Sampat 2010). Additionally, examiner citations are more closely tied to indicators of economic patent value. Empirical evidence has shown that citations by examiners, who evaluate a patent application for patentability, have a higher correlation with value compared to applicant citations (Hegde and Sampat 2009). Despite the improvement of examiner citations over applicant citations, concerns persist that many examiner citations may not be directly linked to the value of the underlying patent.

This paper aims to empirically test a metric that has not been widely used by scholars but may also be tied to private value—the substantive use of a patent by an examiner in rejecting another pending patent application. This analysis focuses on rejections that are based on arguments such as the invention claimed in the patent application is not novel (often identified as “102” based on the statute, 35 U.S.C. § 102) or is obvious (often identified as “103”, 35 U.S.C. § 103). Additionally, analogous information from EPO search reports is examined. Unlike applicant or examiner citations, which may or may not lead to rejections, the patents used in rejections are considered by examiners to disclose or suggest the claims in the patent application, thus establishing a close relationship with subsequent inventions. These rejections result from substantive analysis by the patent office.¹ The theory is that these patent uses, by their very nature, are both more legally and economically relevant than applicant or examiner citations. Patent rejections are not directly influenced by potential strategic behavior from patent lawyers or patent applicants, providing them a more direct connection to private value. It is worth noting that the previous use of rejection citation data has been limited to comparisons between the United States and European patent offices (Wada 2020) and focused on citations before patent issuance when investigating whether such uses are tied to the perceived value of the pending patent application (Sun and Wright 2022).

This paper examines rejection and citation data for patents issued from 1999 to 2007 and links it to value data.² The rejection uses and citations observed for these patents occur over a defined time period—in US patent applications pending from 2008 to 2017. This paper then proceeds in three parts. First, it assesses the extent to which additional information is provided by the use of novelty (§ 102) and obviousness (§ 103) citations compared

to applicant and examiner citations. The overlap between citations made by applicants, examiners, and their use or non-use, in a novelty or obviousness rejection is determined. This information quantifies whether a patent's use in a novelty and obviousness rejection provides additional information beyond who cited the patent. Analogous information from EPO search reports—particularly X and Y citation of US patents from the same period—is also examined. Second, the study explores how patents used in novelty and obviousness rejections (and analogous EPO search reports) relate to common measures of private value—specifically patent renewal, the assertion of a patent in litigation, the number of independent patent claims, and number of patent technological classifications. Our findings show that rejection patents are independently, positively correlated with various of the value measurements beyond applicant citations and examiner citations. The citation as X or Y references in the EPO search report provides similar, independent value information providing a robustness check to the USPTO rejection use findings. By exploring the correlation between rejection citations and these value measurements, we gain insights into the extent to which the use of a patent in rejections, as well as X and Y citation by the EPO, aligns with other indicators of its private value. We also study patents that were both used by the USPTO in rejecting another patent and identified as an X or Y reference in an EPO search report. This comprehensive analysis enables additional understanding of the legal and economic implications of rejection citations, without the traditional reliance on forward citations or applicant citations or even examiner citations.

We then begin the exploration as to whether rejection uses exhibit unique value correlations when compared to applicant or examiner citations. We examine the correlation of rejection uses and applicant and examiner citations to the patent's inclusion in the Orange Book, indicating the patent protects a Federal Drug Administration (FDA) approved drug, or as a Standard Essential Patent (SEP), meaning the patented invention is part of a technological standard. We observe that rejection uses, particularly obviousness rejection uses, are positively correlated with both patent metrics while applicant citations show no statistically significant correlation with either and examiner citations exhibit a negative correlation with Orange Book inclusion. We also study the originality and generality of patents—indicators of the technological dispersion of the patented invention (Trajtenberg, Henderson, and Jaffe 1997). We determined these metric's correlation with citation and rejection uses and find further differences—with applicant citations and obviousness rejection uses being positively correlated with both technology diversity indicators but examiner citations and novelty rejection uses being negatively correlated. These results suggest that rejection uses are more than simply an alternative indicator of value, but provide their own independent insights researchers can leverage.

In summary, this study fills a critical gap in the existing literature by introducing a novel metric that measures a patent's economic value based on its substantive use in rejections by patent examiners as well as X and Y citation by in an EPO search report. By examining the relationship between rejection citations and common measures of private value, we shed light on the legal and economic information provided by rejection citations. This research contributes to the ongoing discourse on patent citations and their implications for the assessment of patent value,

providing insights for both legal and economic analyses in the field of law and economics.

2 | Use of Patents in Rejections

2.1 | Comparing Citation and Patent Rejection Use Data

We use the rejection and citation information from the USPTO Office Action Dataset from 2008 to 2017 (Lu, Myers, and Beliveau 2017). This dataset identifies rejections in office actions issued for patent applications filed over this time period, as well as all citations in those applications. For each office action, the dataset provides the following: every patent number used in the rejection, whether the applicant and/or examiner cited it, and whether it was used in a novelty (§ 102) or obviousness (§ 103) rejection.

We focus on rejections using US utility patents that issued from 1999 to 2007, which consisted of 1,452,454 patents, with these patents being identified using the Patent Examination Research Dataset (PatEx) provided by the USPTO³ (Graham, Marco, and Miller 2015). We limit our analysis to this set of utility patents to ensure the patents issued and were thus available for rejection usage and citation in 2008, the beginning year of the Office Action Dataset. For these patents issued from 1999 to 2007, we examine six measures of importance,

all tabulated for the period between January 2008 and July 2017—the time period covered by the Office Action Dataset: the total number of times the patent was (1) cited by anyone, (2) cited by the applicant, (3) cited by the examiner, (4) used in any rejection; (5) used in a § 102 or novelty rejection, and (6) used in a § 103 or obviousness rejection. Citations and rejections were not double counted if they occurred more than once in the same patent examination process. That is, citations and rejections were observed on a cited/used patent by pending application basis. For example, if an examiner used a patent in multiple novelty rejections during the same examination of a pending application, we counted this as a single novelty rejection for that previously issued patent. This is the same way citations are traditionally counted—one citation is recorded even if the citation is referenced multiple times by the citing reference.

We therefore take the universe of US utility patents issued from 1999 to 2007 and then observe how these issued patents are cited and used in rejections in office actions in other, patent applications pending from 2008 to 2017. This is depicted on a timeline below in Figure 1.

The following table reports the distribution of who cited these patents and the number of times they were used in novelty and obviousness rejections. Table 1 shows that there are far fewer examiner citations as compared to applicant citations, with nearly four times as many examiner citations on average, and

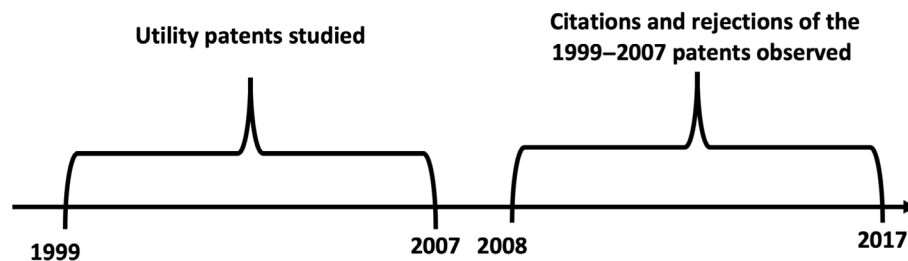


FIGURE 1 | Patents included in the present study.

TABLE 1 | Distribution of citations and rejection uses for all US utility patents issued from 1999 to 2007.

	All citations	All applicant citations	All examiner citations	All rejection uses	Novelty rejections uses	Obviousness rejections uses
Mean	6.478	5.157	1.756	0.840	0.248	0.760
Std. dev.	14.463	13.427	2.854	1.620	0.662	1.505
10th %	0	0	0	0	0	0
25th %	1	0	0	0	0	0
Median	2	1	1	0	0	0
75%	7	5	2	1	0	1
90th %	15	12	5	2	1	2
Min	0	0	0	0	0	0
Max	1183	1179	244	159	97	141
N	1,452,454	1,452,454	1,452,454	1,452,454	1,452,454	1,452,454

that rejection uses, and specifically novelty and obviousness rejection uses, are even fewer, with the means below 1 and 90th percentile beginning with only one novelty rejection and two obviousness rejections.

Of the dataset of all US utility patents issued from 1999 to 2007, there are 972,846 patents cited at least once by the applicant, 876,895 patents cited at least once by the examiner, 261,567 patents used at least once in a novelty rejection, and 566,729 patents used at least once in an obviousness rejection.

The patents frequently used in rejections were not necessarily the same as the patents with the highest forward citations, applicant citations, or examiner citations. The Sankey diagram in Figure 2 shows the relationship between all forward citations and novelty and obviousness rejection uses combined. Figure 3 shows the relationship between all examiner citations and such rejection uses. In both figures, there is a shuffle amongs quantiles (labeled Q1–Q4) from the citation of a patent, even by an examiner, to its use in a rejection. Even patents in the lower quantiles of all citation and examiner citation metrics appear in higher quantiles of actual rejections, and vice versa, emphasizing the new information that rejection use provides.

We also observed, on a pending application basis, when the various citations and rejection uses occurred on their own or in combination. For example, we noted when a patent was cited by an applicant during an application's examination, but was not cited by the examiner nor used in a rejection during that same application's examination. This allowed us to observe the overlap between citations and rejection uses during the same examination. Accordingly, rejection uses of a patent never appeared by themselves—these patents were also either cited by the applicant and/or cited by the examiner. However, not all applicant citations or examiner citation coincided with a rejection use. Roughly 90% of rejection uses were also examiner citations (1,066,418, 87.43%), with a small percentage of rejection uses coming exclusively from applicant citations (153,285, 12.57%). The Venn diagram in Figure 4 below shows the overlaps between a patent issued in 1999–2007 being cited by an applicant, an examiner, or used in a rejection in applications pending between 2008 and 2017.

The distribution of exclusive citations and rejection uses in our dataset are reported in Table 2 below. Notably, the mean, standard deviation, and distribution of examiner citations *not* cited by the applicant and *not* used in a rejection is very similar to that of rejection uses.⁴

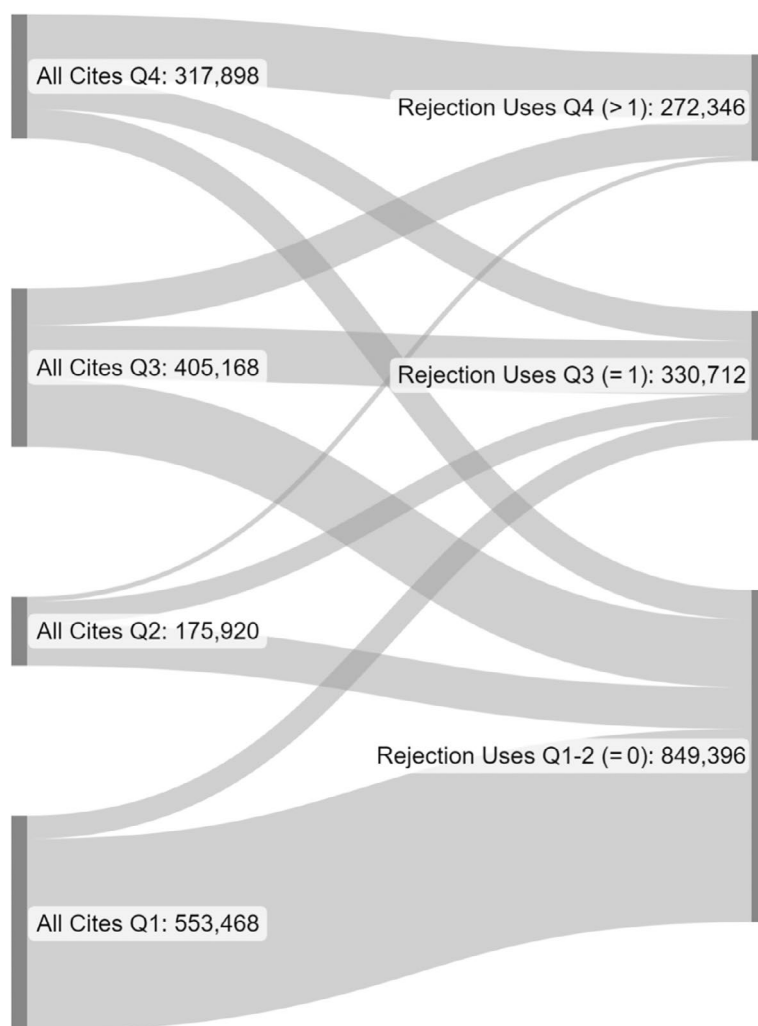


FIGURE 2 | All forward citations compared to all rejection uses.

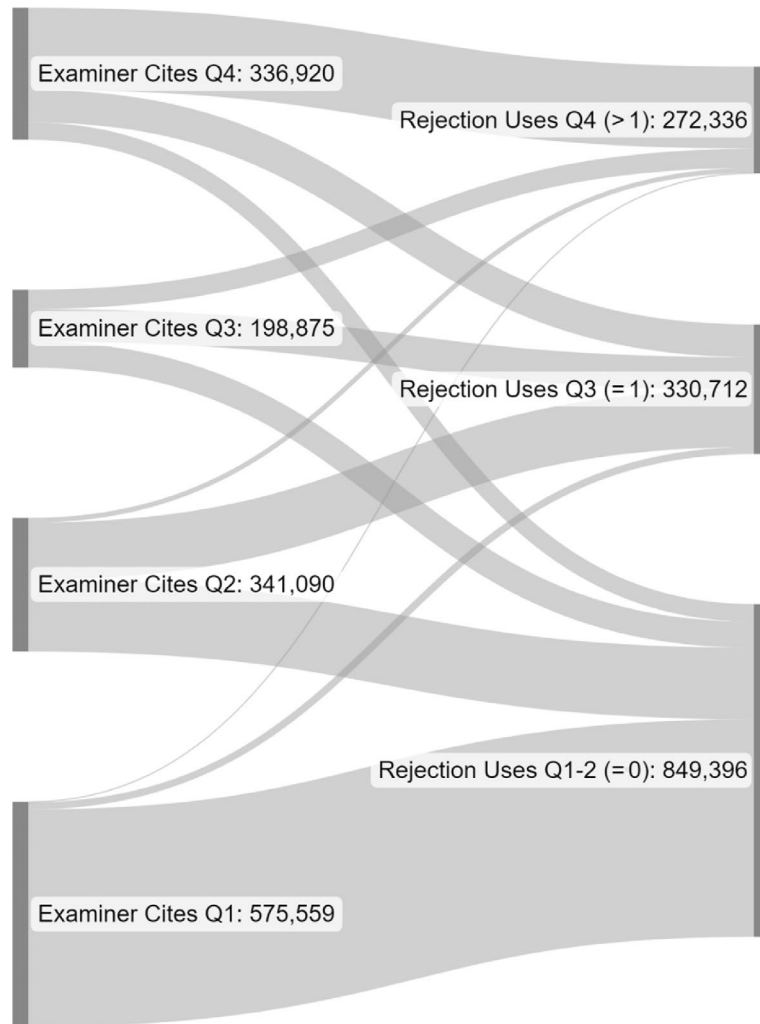


FIGURE 3 | All examiner citations compared to all rejection uses.

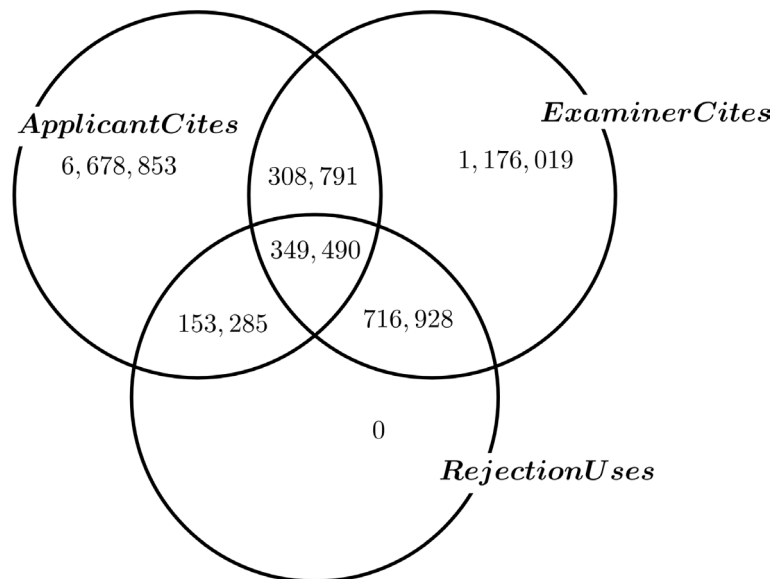


FIGURE 4 | Venn diagram of applicant citations, examiner citations, and rejection uses.

2.2 | Comparing Citation and EPO Search Report Data

European Patent Office search report citation provides an analogous counterpart to USPTO rejection usage. In EPO search reports, US patents are often cited and the level of relevance of those patents to the patent application, typically pending before the EPO, is classified by the EPO into various categories. An EPO citation indicated by the letter “X” indicates “that document is such that when taken alone, a claimed invention cannot be considered novel” (European Patent Office 2023). And a category “Y” citation indicates “that a document is such that a claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other documents of the same category, such combination being obvious to a person skilled in the art.” (Id.). EPO search reports also cite references in categories not

bringing the patent application’s patentability into question, such as an “A” category citation indicating the cited reference is merely background for the state of the art and does not bring the patent application’s novelty or inventive step into question. Accordingly, as other scholars have observed (Wada 2020), X and Y citations in EPO search reports closely resemble 102 and 103 rejections by the USPTO. There are differences. USPTO rejections are actual rejections from the USPTO to pending US patent applications, while EPO search report X and Y references suggest that a claim of a pending patent application, typically filed in the EPO, will be rejected unless the claims are narrowed. Furthermore, although the concept is essentially the same, the standard for obviousness under US patent law is not worded identically as the EPO’s examination for whether the claimed invention exhibits an “inventive step.” (Id.).

Given the similarity, we can examine whether X and Y citations in EPO search reports offer comparable insights into patent value, akin to the usage of 102 and 103 rejections. We gathered EPO search report citation information for all US patents (not European patents) in our dataset from the Organisation for Economic Co-operation and Development (OECD) Citations Database (OECD 2023).

The following table reports the distribution of who relied upon these patents and the number of times they were cited in search reports by the EPO as X or Y references. Table 3 shows that there are significantly fewer X and Y citations as compared to both all search report citations and search report citations that are not X or Y citations. X and Y citations are scarce, with the means well below 1 and 90th percentile beginning with only one X citation and no Y citations.⁵ The “All citations” column in Table 3 below include citations in EPO search reports—not only X and Y citations but also A and other categories of citation.

Of the dataset of all US patents issued from 1999 to 2007, there are 387,109 patents identified at least once by the EPO in a search report, 187,906 patents identified at least once as an X reference and 122,399 patents identified at least once as a Y reference. This comparative distribution between X and Y references is different than that exhibited in USPTO novelty and obviousness

TABLE 2 | Exclusive distribution of citations and rejection uses for all US utility patents issued from 1999 to 2007.

	Applicant citations not used in rejections	Examiner citations not cited by applicant and not used in rejection	Rejection uses
Mean	4.811	0.810	0.840
Std. dev.	13.013	1.486	1.620
10th %	0	0	0
25th %	0	0	0
Median	1	0	0
75th %	4	1	1
90th %	12	2	2
Min	0	0	0
Max	1150	78	159
N	1,452,454	1,452,454	1,452,454

TABLE 3 | Distribution of citation categories by EPO in search reports for all US patents issued from 1999 to 2007.

	All citations	Citations not X or Y	X citations	Y citations
Mean	0.468	0.223	0.176	0.104
Std. dev.	1.091	0.632	0.549	0.391
10th %	0	0	0	0
25th %	0	0	0	0
Median	0	0	0	0
75th %	1	0	0	0
90th %	2	1	1	0
Min	0	0	0	0
Max	52	48	28	33
N	1,452,454	1,452,454	1,452,454	1,452,454

rejections. The differences may be due to these being in a search report, while the USPTO uses are in actual rejections, and the substantive differences between the obviousness standard under US patent law and inventive step test in Europe (Wada 2020). It may also be that the USPTO is likely to combine more references when issuing an obviousness rejection, while the EPO uses fewer in its combinations.

2.3 | Patent Valuation Data

We now compare our novel patent rejection metric to existing measures of private patent value. Here we are looking at the value metrics for all US patents issued between 1999 and 2007. First, we look at patent renewals—the payment of maintenance fees to keep the patent enforced that come due at the end of the 4th, 8th, and 12th year after a patent issue. Non-payment causes the patent's exclusivity to lapse. Studies have relied upon maintenance fee payments to indicate a patent's private value (Pakes 1986; Hegde and Sampat 2009; Lanjouw, Pakes, and Putnam 1998). Since renewals are accepted in the literature as a measure of value, we benchmark our new patent rejection use, and X and Y citation by the EPO, measure to renewals. We collected patent renewal data for all patents issued from 1999 to 2007. This data came from the USPTO maintenance fee events data set, available via the USPTO Open Data Portal and includes all recorded maintenance fee events for every patent granted from September 1, 1981 to present.⁶ We then encode the renewal data to a single variable that indicates what stage of renewal a patent reaches. The renewal variable ranged from 0, equating to never renewed, to 1 (only renewed once at year 4), 2 (renewed up to year 8), and 3 (fully renewed all the way through year 12). A summary of this renewal data is set forth in Tables 4 and 5 below.

Second, we look at patent litigation—the filing of a suit to assert a patent's exclusivity against another company or individual—another common measure of private value (Harhoff, Scherer, and Vopel 2003). We use data from the USPTO Patent Litigation Docket Reports⁷ including all United States district court patent litigation from 1963 to 2020, including more than 158,165 suits involving 56,489 patents (Toole, Miller, and Sichelman 2024). A summary of the litigated patents in our dataset is set forth below in Table 5.

Third, we captured a variety of patent characteristics in the cited patent that can also be an indicator of private value (Lanjouw and Schankerman 2004; Marco, Sarnoff, and deGrazio 2019). These are characteristics of the patent used as prior art, not in the patent application that was rejected. This data was obtained from PatentsView⁸ and the PatEx dataset,⁹ both associated with the USPTO and its Office of the Chief Economist (Graham, Marco, and Miller 2015). The number of independent claims for each of the patents in our dataset was obtained and is described below in Table 5. In addition, the number of CPC patent classifications for each patent are reported in Table 5. Patent classifications categorize the technologies in the patent (Lafond and Kim 2019).

3 | Patent Value, Citation, and Rejection Uses

To determine whether substantive use of a patent by an examiner in a rejection correlates with the various valuation metrics,

TABLE 4 | Percentage of all US utility patents issued 1999–2007 that reach each renewal stage.

Renewed at 4	Renewed at 8	Renewed at 12
0.871	0.685	0.476

we performed a series of regressions with the measures of use in a rejection. The purpose of this section is to test whether rejection uses correlate positively with commonly used patent value metrics.

3.1 | Patent Value and USPTO Rejection Uses

For the renewal, litigation, number of independent claims, and the number of CPCs value metrics, we performed a linear regression with the errors clustered at the examiner art unit level.¹⁰ We also, with each linear regression, controlled for the examiner art unit and year of the cited patent by using a combined, examiner art unit and issued year fixed effect.¹¹ While we report only one model for each value metrics below for readability purposes, a variety of other models were performed as well and produced similar results.¹²

The model reported in Table 6 below uses independent variables for both applicant citations and examiner citations that are both (a) mutually exclusive of each other and (b) of the novelty and obviousness rejection uses variables. The applicant citations used in the model are only those applicant citations *not* used in rejections. In addition, the examiner citations used in the model reported below are only those *not* used in rejections and *not* appearing in applicant citations. We do this to avoid multicollinearity, which if present can make it difficult to determine the individual effect of each independent variable, in particular the rejection use variables. Multicollinearity is a particular problem here where, as graphically depicted in Figure 4, all rejection uses appear in either examiner or applicant citations, with an overwhelming majority appearing in examiner citations.¹³

Based on the positive, statistically significant correlations, rejection uses are a viable alternative to other citation-based value proxies. We find that obviousness rejections have a statistically significant, positive relationship to all value metrics. Rejections for novelty have a statistically significant, positive relationship to all value metrics except being litigated, where statistical significance was lacking in the model reported.¹⁴ Of course, our regressions do not show causation, merely correlation.

Novelty and rejection uses correlate with greater private value. For example, the coefficients of the novelty and obviousness rejection use variable indicate a 2.12, for novelty rejection uses, to 3.03, for obviousness rejection uses, percentage point increase, on average, in the renewal metric. This means that each novelty and obviousness rejection correlates with the patent being closer to the next renewal stage—the next maintenance fee being paid. Given, as shown in Table 3, the distribution of examiner citations not cited by the applicant or used in a rejection is similar to the distribution of rejection uses, one can compare the examiner

TABLE 5 | Summary statistics on renewal, litigation, independent patent claims, and technology classes for all US patents issued 1999–2007.

Patents issued from 1999 to 2007	Renewal	Litigated?	Number of independent claims	Number of CPC classes
Mean	2.0307	0.014	3.015	5.495
Std. dev.	1.0864	0.119	2.540	6.579
10th %	0	0	1	1
25th %	1	0	1	2
Median	2	0	2	4
75th %	3	0	4	7
90th %	3	0	6	11
Min	0	0	1	1
Max	3	1	130	331

TABLE 6 | Linear regression on patent value metrics.

	Renewal	Litigated	Number of independent claims	Number of CPC classes
Applicant citations not used in rejection	0.006** (0.000)	0.001** (0.000)	0.009** (0.001)	0.056** (0.004)
Examiner citations not cited by applicant and not used in rejection	0.009** (0.002)	0.001** (0.000)	0.050** (0.004)	0.170** (0.022)
Novelty rejection uses	0.021** (0.003)	0.000 (0.000)	0.029** (0.007)	0.119** (0.025)
Obviousness rejection uses	0.030** (0.002)	0.001** (0.000)	0.061** (0.004)	0.148** (0.016)
Constant	1.963**	0.009**	2.878**	4.944**
R^2	0.102	0.020	0.062	0.133

Note: We control for the year the cited patent was issued and the examiner art unit of the cited patent. Standard errors clustered at examiner art unit level are reported in parentheses.

* $p < 0.05$.

** $p < 0.01$.

citation coefficient to the rejection coefficients. In addition, the examiner coefficient is smaller, 0.009, or roughly 1%, on average, of a renewal level compared with the 2% to 3%, on average, for rejection uses.¹⁵

3.2 | Patent Value and EPO Search Report Citations

As a robustness check, we performed a similar analysis to investigate whether EPO search report X or Y citation also positively correlate to the same patent value metrics.¹⁶ We performed the identical regressions, but instead of focusing on USPTO rejection usage, our attention shifted to the EPO search report counterpart—X and Y citations—of the US patents within our dataset. Notably, the value perspective in EPO search reports differs slightly since we are examining whether the usage by a foreign patent office provides insights on a US patent. As we did with Table 6, the model reported below avoids multicollinearity

by introducing an independent variable that includes only EPO search report citations that are not X or Y citations.¹⁷

As with USPTO rejection usage, both X and Y citations have a statistically significant, positive relationship to all value metrics, even after controlling for non-X and Y citation measures and the examiner art unit and issue year of the patent.

We next sought to determine whether a patent's value became even more pronounced if that patent was both used in a USPTO rejection and categorized as an X or Y EPO citation. These patents—utilized in rejecting another US patent and appearing as an X or Y reference in an EPO search report—may be perceived of as having greater relevance to the state of invention, not only in the United States but also globally. To test this hypothesis, we analyzed renewal rates for three categories of patents in our dataset—those used in only USPTO rejections, those cited as only X or Y references in EPO search reports, and those employed both in USPTO rejections and cited as X or Y references.

Of the US patents that were issued from 1999 to 2007, 449,917 of them were used in only USPTO rejections, 94,020 were exclusively cited as X or Y references by the EPO, and 153,141 were both used in USPTO rejections and categorized as a X or Y reference Table 7.

The results of linear regressions on the patent value metrics previously used—renewal, litigation, number of independent claims, and number of CPC classes—is reported in Table 8 below.

Patents used by both the USPTO and EPO exhibited a positive, statistically significant correlation in all value categories. This analysis suggests that combining the information from both patent offices can provide another indicator of a patent's value. The concurrent validation of the patent's high relevance to patent applications pending in both offices appears to correlate highly with whether that patent is renewed, litigated, has more independent claims, and appear in more patent classifications and, thus, is of private value to its owner.

4 | Discussions and Conclusion

Based on the above analysis, the use of patents in rejections and categorization as X or Y references is a viable alternative value metric to the currently used applicant and/or examiner citations. In addition, this alternative can prove particularly useful in some situations given that there are patents whose citation, even by examiners, is not correlated with usage in novelty or obviousness rejections or inclusion in an EPO search report as an X or a Y citation. The use of patents in rejections provides additional legal information that does not completely overlap with applicant or examiner citation information. As visually demonstrated in Figures 2, 3 and 4 above, use in rejections can provide unique legal information that is a validated proxy for patent value. In addition, when combining the information about a given patent from both the USPTO and EPO, insights about that patent's value are also obtained.

As with previous projects, such as Hegde and Sampat (2009), we aim to document this regularity so that other researchers can

TABLE 7 | Linear regression on value metrics for EPO search report citations.

	Renewal	Litigated	Number of independent claims	Number of CPC classes
Search report citations not X or Y	0.083** (0.004)	0.004** (0.000)	0.101** (0.005)	0.460** (0.018)
X citations	0.056** (0.003)	0.003** (0.000)	0.121** (0.006)	0.626** (0.027)
Y citations	0.063** (0.004)	0.002** (0.000)	0.094** (0.007)	0.468** (0.026)
Constant	1.996**	0.013**	2.961**	5.234**
R ²	0.096	0.013	0.056	0.120

Note: We control for the year the cited patent was issued and the examiner art unit of the cited patent. Standard errors clustered at examiner art unit level are reported in parentheses.

* $p < 0.05$.

** $p < 0.01$.

TABLE 8 | Linear regression models on value metrics and usage by EPO and USPTO.

	Renewals	Litigated	Number of independent claims	Number of CPC classes
Only used in USPTO rejections	0.173** (0.006)	0.008** (0.000)	0.280** (0.009)	0.770** (0.043)
Only cited as X or Y references by EPO	0.134** (0.006)	0.003** (0.000)	0.169** (0.011)	0.842** (0.037)
Used in both USPTO rejections and cited as X or Y references by EPO	0.308** (0.009)	0.013** (0.001)	0.518** (0.014)	2.042** (0.075)
Constants	1.936**	0.010**	2.863**	4.987**
R ²	0.010	0.014	0.059	0.120

Note: We control for the year the cited patent was issued and the examiner art unit of the cited patent. Standard errors clustered at examiner art unit level are reported in parentheses.

* $p < 0.05$.

** $p < 0.01$.

leverage rejection uses, as well as X and Y EPO citation's, to obtain greater insight into a patent's value in their studies.

The results are, notably, intuitive given that novelty and obviousness rejections and X and Y citation are by definition directly tied to the subject matter of the citing patent application. By using the patent in a rejection or categorizing as an X or Y citation, the examiner is indicating that the knowledge in the used patent is either being used again, in a novelty rejection, or being minorly improved, in an obviousness rejection. The rejection gives us direct evidence that a firm both applied for, and incurred costs to attempt to obtain, a patent on the knowledge already contained in the used patent. Patent rejection metrics are not influenced by strategic behavior by the patent applicant or patent attorney. The same holds for X or Y references located in an EPO search report.

The degree of this relationship is also captured in the difference between a novelty (all of the knowledge is disclosed in the used patent) and obviousness (some of the knowledge is disclosed in the used patent) rejection when considering in the context of the prevalence of the two types of rejections (by definition, more patents are used in obviousness rejections, leading to more obviousness rejections when compared to novelty rejections). The results even map these two factors—degree of difference and prevalence.

We mainly leave to future research ways to improve upon novelty and obviousness rejection metrics and find the various, unique information rejection uses may unveil. We did, however, begin exploring what specific information rejection uses may provide when compared to applicant or examiner citations. That is, while both citation and rejection uses all generally correlated positively with value, rejection uses, given their unique nature, may correlate with other patent value metrics that applicant and/or examiner citations do not.

We, therefore, looked at four other patent metrics that can correlate with value. These include whether the patent (or its published application) are a standard essential patent, whether the patent is ever listed in the Orange Book, and the patent's "generality" and "originality" scores. We examine whether rejection uses exhibit different correlations with these value metrics when compared to applicant and examiner citations. We observed some heterogeneity, suggesting there is promise that rejection uses provide value information not captured in applicant or examiner citations.

A patent is considered a SEP if the patent claims an invention essential to the implementation of a particular technology standard. SEPs are most common in the information technology and electronics space and considered valuable to both society—given that a standard can ensure safety, interoperability and compatibility amongs products and/or services—and to the patent holder—where a royalty stream typically accompanies inclusion of the patent in a SEP pool. We used Northwestern University Searle Center's dataset¹⁸ on technology standards and standard setting organization to identify which patents (or their corresponding published applications) in our dataset of all US utility patents issued between 1999 and 2007 are declared

as SEPs (Baron and Pohlmann 2018). A total of 3825 (0.26%) of the patents we studied were identified in this dataset as being declared SEPs.

Patents may also be of value if they claim pharmaceuticals approved by the FDA. There is a public document, called the "Orange Book," that links unexpired patents with small-molecule drugs marketed in the United States (Durvasula et al. 2023). A patent being listed in the Orange Book indicates the patent's exclusivity over a commercialized product, providing an inference that the patent has value both to the patent owner—given the owner's ability to prevent entry of generic competition—and society—with the drug having been approved the FDA for some medicinal purpose. We utilized the open-access dataset of Orange Book records introduced by Durvasula et al. and linked Orange Book listed patents through 2016 with those patents in our 1999–2007 dataset. A total of 1483 (0.10%) patents issued between 1999 and 2007 were listed at one point in time in the Orange Book.

We also computed the "generality" and "originality" of the patents in our dataset as those terms are defined Trajtenberg, Jaffe, and Henderson (Trajtenberg, Henderson, and Jaffe 1997). "Generality" measures the percentage of citations received by a patent that are outside of the patent's designated technological class. That is, the higher the generality percentage, the more forward citations in different technological fields that patent receives than the patent's identification classification (Hall, Jaffe, and Trajtenberg 2001). "Originality" focuses on this differential but for a patent's backwards citations. Originality is computed by measuring the percentage of backward citations a patent makes to references outside of the patent's designated technological classification. (Id.). High generality and high originality scores are viewed by some economists as indicators of a patent's social value since these are patents that include innovations either contributing to and/or drawing from a wide range of technological fields (Baron and Delcamp 2012). We computed the generality and originality, using the formulas identified by Trajtenberg, Jaffe, and Henderson, for the patents in our dataset—all US utility patents issued from 1999 to 2007—using the PatentsView citation and classification information.

Table 9, below, describes the distribution of SEP inclusion, Orange Book listing, generality, and originality percentages for all US utility patents issued from 1999 to 2007.

We then performed a linear regression analysis that is reported in Table 10 below, using the same methodology reported in Table 6 to define the independent variables and substituting the new value metrics as dependent variables.¹⁹ For the Orange Book listing analysis, the regressions were limited to those patents in the drug technology space by leveraging the patent's identification as NBER category 3 or not (Hall, Jaffe, and Trajtenberg 2001), given that patents in other technological areas very unlikely to be Orange Book listed regardless of citations or rejection uses. For the standards essential patents analysis, the regressions are limited to patents in the computers and electrical space—NBER categories 2 and 4—since these are the main technological spaces where standards essential patents are commonly found.

TABLE 9 | Distribution of additional value metrics for all utility US patents issued from 1999 to 2007.

Patents issued from 1999 to 2007	SEP inclusion	Orange Book listing	Generality	Originality
Mean	0.003	0.001	0.582	0.605
Std. dev.	0.051	0.032	0.316	0.285
10th %	0	0	0.111	0.389
25th %	0	0	0.333	0.196
Median	0	0	0.600	0.637
75th %	0	0	0.864	0.851
90th %	0	0	1	0.973
Min	0	0	0	0
Max	1	1	1	1

TABLE 10 | Linear regression models on additional value metrics.

	SEP inclusion (NBER = 2 or 4)	Orange Book listing (NBER = 3)	Generality	Originality
Applicant citations not used in rejection	0.0000 (0.0000)	0.0000 (0.0000)	0.0013** (0.0002)	0.0009** (0.0001)
Examiner citations not cited by applicant and not used in rejection	0.0003* (0.0001)	−0.0014* (0.0006)	−0.0017* (0.0007)	−0.0015** (0.0005)
Novelty rejection uses	−0.0003 (0.0001)	−0.0008 (0.0005)	−0.0138** (0.0011)	−0.0091** (0.0008)
Obviousness rejection uses	0.0005* (0.0001)	0.0025** (0.0008)	0.0044** (0.0008)	0.0029** (0.0005)
Constant	0.0050**	0.0068**	0.5774**	0.6015**
R ²	0.0384	0.0283	0.1130	0.0638

Note: We control for the year the cited patent was issued and the examiner art unit of the cited patent. Standard errors clustered at examiner art unit level are reported in parentheses.

* $p < 0.05$.

** $p < 0.01$.

As reported in Table 10, the correlation between rejection uses and these additional value metrics does not perfectly mirror the correlations with applicant or examiner citations. For example, while the correlation between applicant citations and novelty rejection uses and SEP inclusion are not statistically significant, there is a statistically significant relationship for examiner citations and obviousness rejection uses. And the directions of the correlation are different—with examiner citations negatively correlated with SEP inclusion while obviousness rejection uses are positively correlated. A varying relationship is also observed with Orange Book listings, with examiner citations being negatively correlated, while obviousness rejection uses are positively correlated. In addition, both applicant citations and novelty rejection uses show no statistical significance to Orange Book listings. The patent's generality and originality also exhibit this difference in relationship between examiner citations and obviousness rejection uses—with the coefficients signs flipping. And interestingly, novelty rejection uses are also negatively correlated with generality and originality.

While we do not fully investigate the mechanisms for such heterogeneities, the more limited instances of rejection uses may make them better equipped to correlate with other extremely limited occurrences such as SEP inclusion or Orange Book listings. That is, SEP and Orange Book inclusion is so rare, applicant or examiner citation fail to provide enough information to identify these value metrics. Rejections, on the other hand, provide more exacting information about the patents' place within the technological space. And obviousness rejection uses in particular—which means the later application is building upon the earlier patent, not simply duplicating it—lines up with a patent disclosing either an “essential” technology that is widely used by others (i.e., a SEP) or a commercially available drug (i.e., Orange Book listed) that others find value in and are trying to find different uses for or make variations of. The generality and originality findings fit a similar line of reasoning—with examiner citation and novelty rejection uses being technologically narrow—applicable to a specific application within a specific examiner art unit—while obviousness rejection uses are more

likely to go beyond one particular technological field because they disclose a part, but not all of, a pending application's claim.

The above is just a small exploration into using rejections as more than mere alternatives to applicant or examiner citations. Rejection uses can provide patent value information by themselves. And we believe that rejection use metrics can be refined even further by isolating (1) patents used in “final” rejections, as compared to non-final rejections, (2) patents that blocked all claims in an application rather than only some claims; (3) patents used in multiple rejections during the course of patent office consideration of a single application, and (4) patents that could not be overcome by the applicant and which led to abandonment of the application. Such refinements may provide a whole host of rejection use specific information about a patent that is useful to researchers and analysts.

Another key question to consider is how thoroughly examiners search, and what resources examiners rely on in conducting the search. Toward that end, an open question is how the examiner first learns about a patent when she uses it in a rejection. We proffer four possibilities: (1) the examiner discovered the patent herself in an earlier prosecution (so it should be an examiner-provided reference in a previous patent application); (2) the patent was cited by another applicant in a previous prosecution (so it should be an applicant-provided reference in a previous patent application); (3) the current applicant provided the patent to the examiner in an Information Disclosure Statement, a form used by patent applicants to disclose prior art to the USPTO, and the examiner had not previously seen it in a past prosecution; and (4) the examiner found the patent herself and had not seen it in a past prosecution. Future researchers may categorize which of these possibilities rejection patents fall into. The fourth possibility, for example, might suggest the examiner actually conducted a specific search for the patent as compared to merely relying on a stock patent that she was familiar with from a previous application.

In sum, our findings reveal that rejection uses, as well as X and Y EPO search citations, independently exhibit positive correlations with most of the commonly used value measurements. In addition, rejection uses even show promise to provide patent value information beyond that exhibited by applicant and examiner citations. Thus, rejection use metrics shows promise of being an independent tool for evaluating a patent's value, making rejection uses worthy of further research and exploration.

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Data Availability Statement

Data necessary to replicate the results of this article are available upon request from the corresponding author. The data have also been uploaded to the Harvard Dataverse.

Endnotes

- ¹ This potential tighter relationship is also supported by studies looking at EPO identification of patent citations as X or Y references in search reports (Criscuolo and Verspagen 2008). However, EPO search reports do not perform the same substantive analysis as a rejection nor do they, by their citation, prevent the issuance of a patent application.
- ² The present study focuses on citations to US patents and their published patent application counterparts. In office actions, there are also citations to published patent applications that do not result in issued patents and non-patent literature. For a discussion of the use of abandoned patent applications as the basis for rejections, (Cotropia and Schwartz 2020). We combine citations to U.S. patents and their published patent applications, but we do not combine or group citations by US patent family such as continuations or divisionals.
- ³ This dataset is available at <https://www.uspto.gov/ip-policy/economic-research/research-datasets/patent-examination-research-dataset-public-pair>.
- ⁴ A correlation matrix in Table 1A of the Online Appendix provides further description of the distribution of citation and rejection uses on a patent-by-patent basis.
- ⁵ A correlation matrix in Table 2A of the Online Appendix provides further description of the distribution of citation and rejection uses on a patent-by-patent basis.
- ⁶ Available from <https://developer.uspto.gov/product/patent-maintenance-fee-events-and-description-files>. This file is updated weekly. We used the event data set released on April 20, 2024 for this study.
- ⁷ This dataset is available from <https://www.uspto.gov/ip-policy/economic-research/research-datasets/patent-litigation-docket-reports-data>.
- ⁸ This dataset is available from <https://patentsview.org/>.
- ⁹ This dataset is available from <https://www.uspto.gov/ip-policy/economic-research/research-datasets/patent-examination-research-dataset-public-pair>.
- ¹⁰ We perform linear regressions for two additional value metrics in the Online Appendix—total number of claims and minimum number of words in the independent claims. These results are reported in Table 11B of the Online Appendix.
- ¹¹ We also performed all of the linear regressions with a CPC and year fixed effect, instead of art unit. The result is reported in Table 7B of the Online Appendix.
- ¹² We performed the analysis modifying the independent variables—specifically either adding 1 to the citation and rejection use independent variables or taking the log of those variables after 1 is added. The results of this analysis are reported in Tables 6B and 7B of the Online Appendix. We also performed the analysis on an individual, issued year basis, which is reported in Table 8B. We further performed an ordered logistic regression for renewal and logistic regression for litigation, reported in Tables 8B and 9B. As fully reported in the Online Appendix, the results are similar—positive correlation of rejection uses to these patent value metrics—to those reported in Table 6 in this paper.
- ¹³ We report other models in the Online Appendix, reported in Tables 1B–4B, that include multicollinearity and rejection uses are still positively correlated with the value metrics.
- ¹⁴ As reported in the Online Appendix Tables 1B–4B, novelty rejection uses exhibited statistical significance, positive relationship in other models and never exhibited a negative relationship with litigation in any of the models.
- ¹⁵ A similar coefficient comparison between applicant citations and rejection uses is not proper given, as reported in Table 3, applicant citations exhibit a very different distribution than rejection uses.
- ¹⁶ In the Online Appendix, we performed similar analysis for two additional patent value metrics—total number of patent claims and

number of words in the shortest independent claim. The results are in Tables 16B and 17B.

¹⁷ The Online Appendix, in Tables 12B–15B, reports a variety of other models that include all EPO search report citations and the results for X and Y citations is still positive and significant for the various patent value metrics.

¹⁸ Available at <https://www.law.northwestern.edu/research-faculty/clbe/innovationeconomics/data/technologystandards/>.

¹⁹ And just as with the earlier regressions in Part 3 of the paper, the additional linear regression models and logistic regression models for SEP and Orange Book inclusion are reported in the Online Appendix, specifically Tables 1C–5C. The results are consistent with those reported in Table 10.

References

- Abrams, D., U. Akcigit, and J. Grennan. 2019. “Patent Value and Citations: Creative Destruction or Strategic Disruption?” https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2351809.
- Alcácer, J., M. Gittelman, and B. Sampat. 2009. “Applicant and Examiner Citations in U.S. Patents: An Overview and Analysis.” *Research Policy* 38: 415–427.
- Arora, A., S. Bazenton, and H. Lee. 2018. “Reversed Citations and the Localization of Knowledge Spillovers.” *Journal of Economic Geography* 18, no. 3: 495–521.
- Baron, J., and H. Delcamp. 2012. “The Private and Social Value of Patents in Discrete and Cumulative Innovation.” *Scientometrics* 90, no. 2: 581–606.
- Baron, J., and T. Pohlmann. 2018. “Mapping Standards to Patents Using Declarations of Standard-Essential Patents.” *Journal of Economics and Management Strategy* 27, no. 3: 504–534.
- Cotropia, C. A. 2009. “Modernizing Patent law’s Inequitable Conduct Doctrine.” *Berkeley Technology Law Journal* 24: 723.
- Cotropia, C. A., M. A. Lemley, and B. Sampat. 2013. “Do Applicant Patent Citations Matter?” *Research Policy* 42, no. 4: 844–854.
- Cotropia, C. A., and D. L. Schwartz. 2020. “The Hidden Value of Abandoned Applications to the Patent System.” *Boston College Law Review* 61: 2809.
- Cotter, T. F. 2011. “An Economic Analysis of Patent Law’s Inequitable Conduct Doctrine.” *Arizona Law Review* 53: 735–779.
- Criscuolo, P., and B. Verspagen. 2008. “Does It Matter Where Patent Citations Come From? Inventor vs. Examiner Citations in European Patents.” *Research Policy* 37, no. 10: 1892–1908.
- Durvasula, M., C. S. Hemphill, L. L. Ouellette, B. Sampat, and H. L. Williams. 2023. “The NBER Orange Book Dataset: A User’s Guide.” *Research Policy* 52, no. 7: 104791.
- European Patent Office. 2023. “Guidelines for Examination in the European Patent Office.” <https://www.epo.org/en/legal/guidelines-epc>.
- Graham, S., A. Marco, and R. Miller. 2015. “The USPTO Patent Examination Research Dataset: A Window on the Process of Patent Examination.”
- Hall, B., C. Helmers, and G. von Graevenitz. 2021. “Technology Entry in the Presence of Patent Thickets.” *Oxford Economic Papers* 73, no. 2: 903–926.
- Hall, B., A. Jaffe, and M. Trajtenberg. 2001. “The NBER Patent Citations Data File: Lessons, Insights and Methodological Tool.” <https://www.nber.org/papers/w8498>.
- Harhoff, D., F. M. Scherer, and K. Vopel. 2003. “Citations, Family Size, Opposition and the Value of Patent Rights.” *Research Policy* 32, no. 8: 1343–1363.
- Hegde, D., and B. Sampat. 2009. “Examiner Citations, Applicant Citations, and the Private Value of Patents.” *Economics Letters* 105, no. 3: 287–289.
- Jaffe, A., and G. de Rassenfosse. 2019. “Patent Citation Data in Social Science Research: Overview and Best Practices.” In *Research Handbook on the Economics of Intellectual Property Law*, edited by B. de Poorter, P. Menell, and D. Schwartz, vol. 2. Cheltenham, UK: Edward Elgar Publishing.
- Jaffe, A., and M. Trajtenberg. 2002. *Patents, Citations, and Innovations: A Window on the Knowledge Economy*. Cambridge: MIT Press.
- Kuhn, J., K. Younge, and A. Marco. 2020. “Patent Citations Reexamined.” *RAND Journal of Economics* 51, no. 1: 109–132.
- Kuhn, J. M. 2010. “Information Overload at the U.S. Patent and Trademark Office: Reframing the Duty of Disclosure in Patent Law as a Search and Filter Problem.” *Yale Journal Law & Technology* 13: 89–140.
- Lafond, F., and D. Kim. 2019. “Long-Run Dynamics of the US Patent Classification System.” *Journal of Evolutionary Economics* 29, no. 2: 631–664.
- Lanjouw, J. O., A. Pakes, and J. Putnam. 1998. “How to Count Patents and Value Intellectual Property: The Uses of Patent Renewal and Application Data.” *Journal of Industrial Economics* 46, no. 4: 405–432.
- Lanjouw, J. O., and M. Schankerman. 2004. “Patent Quality and Research Productivity: Measuring Innovation With Multiple Indicators.” *Economic Journal* 114, no. 495: 441–465.
- Lu, Q., A. Myers, and S. Beliveau. 2017. “USPTO Patent Prosecution Research Data: Unlocking Office Action Traits.”
- Marco, A., J. Sarnoff, and C. deGrazio. 2019. “Patent Claims and Patent Scope.” *Research Policy* 48, no. 6: 1–17.
- Marco, A. C. 2007. “The Dynamics of Patent Citations.” *Economics Letters* 94, no. 2: 290–296.
- Nelson, A., A. Earle, J. Howard-Grenville, J. Haack, and D. Young. 2014. “Do Innovation Measures Actually Measure Innovation? Obliteration, Symbolic Adoption, and Other Finicky Challenges in Tracking Innovation Diffusion.” *Research Policy* 43, no. 6: 927–940.
- OECD. 2023. “Patent Datasets Raw data, STI Micro-Data Lab.” <https://www.oecd.org/sti/inno/intellectual-property-statistics-and-analysis.htm>.
- Pakes, A. 1986. “Patents as Options: Some Estimates of the Value of Holding European Patent Stocks.” *Econometrica* 54, no. 4: 755–784.
- Sampat, B. N. 2010. “When Do Applicants Search for Prior Art.” *Journal of Law and Economics* 53: 399.
- Sun, Z., and B. Wright. 2022. “Citations Backward and Forward: Insights Into the Patent Examiner’s Role.” *Research Policy* 51: 104517.
- Toole, A., R. Miller, and T. M. Sichelman. 2024. “Technical Documentation for Patent Litigation Docket Reports Data, 1963–2020. USPTO Economic Working Paper No. 2024–01 Available at SSRN.” https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4780166.
- Trajtenberg, M. 1990. “A Penny for Your Quotes: Patent Citations and the Value of Innovations.” *RAND Journal of Economics* 21, no. 1: 172–187.
- Trajtenberg, M., R. Henderson, and A. Jaffe. 1997. “University Versus Corporate Patents: A Window on the Basicness of Invention.” *Economics of Innovation and New Technology* 5, no. 1: 19–50.
- von Graevenitz, G., S. Wagner, and D. Harhoff. 2011. “How to Measure Patent Thickets – A Novel Approach.” *Economics Letters* 111, no. 1: 6–9.
- Wada, T. 2020. “When Do the USPTO Examiners Cite as the EPO Examiners? An Analysis of Examination Spillovers Through Rejection Citations at the International Family-to-Family Level.” *Scientometrics* 125: 1591–1615.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.